

# Supplementary Information

Dissecting Sequence, Structure, and Data Recency Effects  
in Enzyme Commission Prediction under Temporal Distribution Shift

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## S1 Related Work

**Classical sequence alignment.** Before machine learning, enzyme function annotation relied primarily on sequence similarity transfer. BLAST [Altschul et al., 1990] identifies homologous proteins by computing pairwise alignment scores against curated databases such as UniProtKB/Swiss-Prot; a query enzyme is annotated with the EC number of its closest hit above an e-value threshold. On the EC-Bench temporal test set, BLASTp achieves a weighted F1 of 0.502 at the 100% similarity threshold, dropping to 0.431 when training is restricted to proteins with at most 30% sequence identity to the test set [Davoudi et al., 2026]—illustrating the well-known fragility of alignment-based methods for evolutionarily distant queries. Profile-based methods such as PRIAM mitigate this limitation through position-specific scoring matrices derived from multiple sequence alignments, but they cannot integrate non-sequence information such as tertiary structure.

**Deep learning for EC prediction.** DeepEC [Ryu et al., 2019] and DEEPRE introduced 1D CNNs over one-hot amino acid sequences with global average pooling; while effective for EC Levels 1–3, these models struggle at Level 4 due to fine-grained specificity requirements and limited receptive fields. ECPick [Han et al., 2024] augments a hierarchical 1D CNN with Dempster–Shafer evidential theory, assigning a trustworthiness score to each prediction. CLEAN [Yu et al., 2023] reframes EC prediction as metric learning: ESM-1b embeddings are projected into a shared latent space trained with a triplet margin loss. HIT-EC [Dumontet et al., 2026] designs a four-level Transformer achieving micro F1 of 0.93 on Swiss-Prot—the current state of the art for sequence-only EC prediction.

**Protein language model encoders.** ESM-2 [Lin et al., 2023] is a family of masked language models—scaling from 8M to 15B parameters—pre-trained on over 250 million sequences from UniRef90. The 650M-parameter variant produces per-residue representations of dimension 1,280; the [CLS] token embedding encodes a global sequence summary.

**Contact map encoding.** A protein contact map is a binary symmetric matrix  $\mathbf{C} \in \{0, 1\}^{N \times N}$ , where  $C_{ij} = 1$  if the  $C\alpha$  distance between residues  $i$  and  $j$  falls below 8 Å [Gligorić et al., 2021]. Contact maps encode tertiary fold topology in a translation- and rotation-invariant form. DeepFRI [Gligorić et al., 2021] treats the contact map as a sparse adjacency matrix and propagates per-residue features through graph convolutional layers. CLEAN-Contact [Ayres et al., 2024] uses ResNet-50 to encode 2D contact maps, concatenating the resulting representation with ESM-2 features before contrastive learning; on New-392 and Price-149 this achieves a 12% and 16% F1 improvement over sequence-only CLEAN.

**EC-Bench temporal evaluation.** EC-Bench [Davoudi et al., 2026] addresses temporal data leakage by using all reviewed Swiss-Prot entries as of February 2018 for training and 468 proteins added by January 2023 as the test set. The primary metric is the weighted F1-score at Level 4, reported at similarity thresholds of 30%–100%.

## S2 Extended Dataset Details

**Data splits.** We constructed two test sets: (i) *random split* (80/10/10) using a fixed random seed; (ii) *cluster split* using MMSeqs2 [Steinegger and Söding, 2017] at 30% sequence identity, following EC-Bench [Davoudi et al., 2026] to evaluate OOD generalization.

**Split integrity checks.** Because Swiss-Prot contains closely related and sometimes identical protein sequences under different accessions, we audited all train/validation/test partitions for accession and exact-sequence overlap. The random split contains no accession overlap across train, validation, and test sets, but it does contain exact sequence duplicates across partitions; therefore, we treat random-split performance as in-distribution evaluation. For sequence-disjoint evaluation, we use the cluster split, where cluster-train, cluster-validation, and cluster-test contain zero accession overlap and zero exact-sequence overlap. Cluster-test results are reported only for models trained on the corresponding cluster-train split.

**Multi-label annotation statistics.** EC annotations are naturally multi-label: a single protein may catalyze multiple reactions and therefore receive multiple Level-4 EC numbers. In the main Swiss-Prot training split, 8,385 proteins (3.88%) contain more than one Level-4 EC annotation, with up to nine Level-4 labels for a single protein. The average Level-4 label cardinality is 0.893 per protein when partial EC annotations are included. These statistics motivate sigmoid outputs and binary cross-entropy rather than a single-label softmax classifier.

## S3 Training Dynamics

Figure 1 and Table 1 summarize best-checkpoint training dynamics for the main ablation models. All models were trained for up to 30 Phase-1 epochs with frozen ESM-2 representations; Contact-EC-Hier was then trained for an additional 20 Phase-2 epochs with partial unfreezing. B3 (contact map only) converges in approximately 10 epochs, reaching val-hard L4 micro F1 comparable to B1 (0.7650 vs. 0.7655). This fast convergence suggests that the contact-map branch learns a compact structural signal rather than merely memorizing sequence-derived labels. Contact-EC-Hier improves beyond both single-modality baselines during Phase 1 and continues to gain after partial unfreezing in Phase 2, reaching 0.8698 on val-hard. The hierarchical sequence-only B2 baseline remains substantially lower throughout training, consistent with the temporal and val-hard underperformance reported in the main text.

Table 1: Best epoch and val-hard L4 micro F1 for each model.

| Model                      | L4 Micro F1 (val hard) | L4 Macro F1 (val hard) | Best Epoch   |
|----------------------------|------------------------|------------------------|--------------|
| B1 (ESM-2, flat FC)        | 0.7655                 | 0.1353                 | 30           |
| B2 (ESM-2, hier. FC)       | 0.4204                 | 0.0383                 | 30           |
| B3 (Contact map only)      | 0.7650                 | 0.1335                 | 10           |
| Contact-EC (flat FC, ours) | 0.8879                 | 0.2294                 | 29           |
| Contact-EC-Hier            | 0.8698                 | 0.2141                 | 20 (Phase 2) |

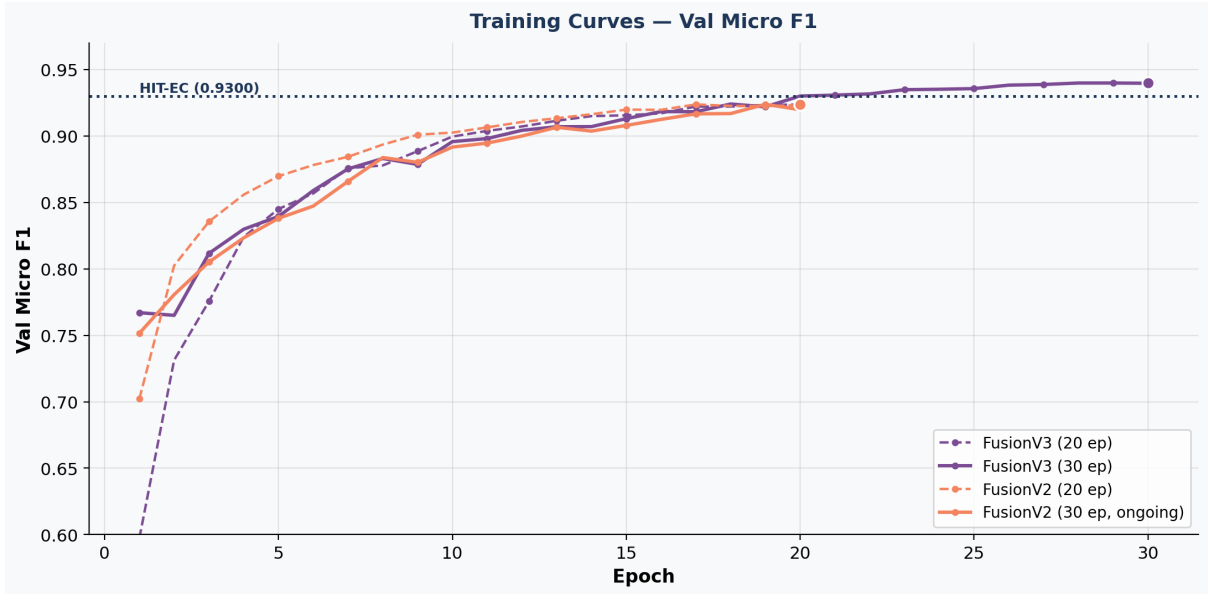


Figure 1: Training dynamics on the EC-Bench hard validation split. Phase 1 uses frozen ESM-2 representations for sequence-containing models; Phase 2 partially unfreezes Contact-EC-Hier. The contact-map-only B3 model rapidly approaches the flat ESM-2 baseline B1, whereas Contact-EC-Hier benefits from sequence–structure fusion and additional Phase 2 optimization.

## S4 Full EC-Bench Leaderboard Comparison

Table S2 reports results on all 468 EC-Bench test proteins, including 309 with partial annotations and 35 with novel EC classes outside the training vocabulary. The all-468 score is reported as an operational closed-set coverage audit, not as a fully defined Level-4 evaluation for partial annotations.

Table 2: Full EC-Bench temporal test: all 468 proteins vs. evaluable 124. The all-468 columns quantify closed-set coverage under label shift; partial-EC proteins are not fully Level-4 scorable, and novel-EC proteins are outside the SP-2018 vocabulary.

| Model                      | All 468       |               | Known-EC 124  |               |            |
|----------------------------|---------------|---------------|---------------|---------------|------------|
|                            | Micro F1      | Weighted F1   | Micro F1      | Weighted F1   |            |
| B1 (ESM-2, flat FC)        | 0.3539        | 0.3156        | 0.4216        | 0.3178        | † EC-Bench |
| B2 (ESM-2, hier. FC)       | 0.1098        | 0.0835        | 0.1111        | 0.0835        |            |
| B3 (Contact map only)      | 0.3167        | 0.2625        | 0.3646        | 0.2734        |            |
| Contact-EC (flat FC, ours) | 0.3716        | <b>0.4763</b> | <b>0.6032</b> | <b>0.5026</b> |            |
| Contact-EC-Hier            | <b>0.3771</b> | 0.4325        | 0.5690        | 0.4528        |            |
| EnzBert-learn†             | –             | 0.463         | –             | –             |            |
| Stacked ensemble†          | –             | 0.524         | –             | –             |            |

leaderboard; weighted F1 on all 468 proteins.

To avoid treating partial EC annotations as Level-4 failures, we additionally evaluate each partial label only up to its deepest annotated hierarchy prefix. For example, EC 3.4.24.- is evaluated through Level 3, whereas EC 1.-.- is evaluated only at Level 1. This prefix-level audit is not directly comparable to complete Level-4 F1, but it quantifies whether the model recovers the known functional hierarchy for incompletely curated proteins.

Table 3: Hierarchical prefix evaluation on the 309 partial-EC temporal proteins. Partial annotations are scored only at levels where ground truth prefixes are available. Level-4 is omitted because no complete Level-4 ground truth exists for these rows.

| Model                      | L1 Micro F1   | L2 Micro F1   | L3 Micro F1   |
|----------------------------|---------------|---------------|---------------|
| B1 (ESM-2, flat FC)        | 0.8311        | 0.5805        | 0.4656        |
| B3 (Contact map only)      | 0.7930        | 0.6615        | 0.5933        |
| Contact-EC (flat FC, ours) | 0.8259        | 0.7322        | 0.6629        |
| Contact-EC-Hier            | <b>0.8409</b> | <b>0.7685</b> | <b>0.6975</b> |

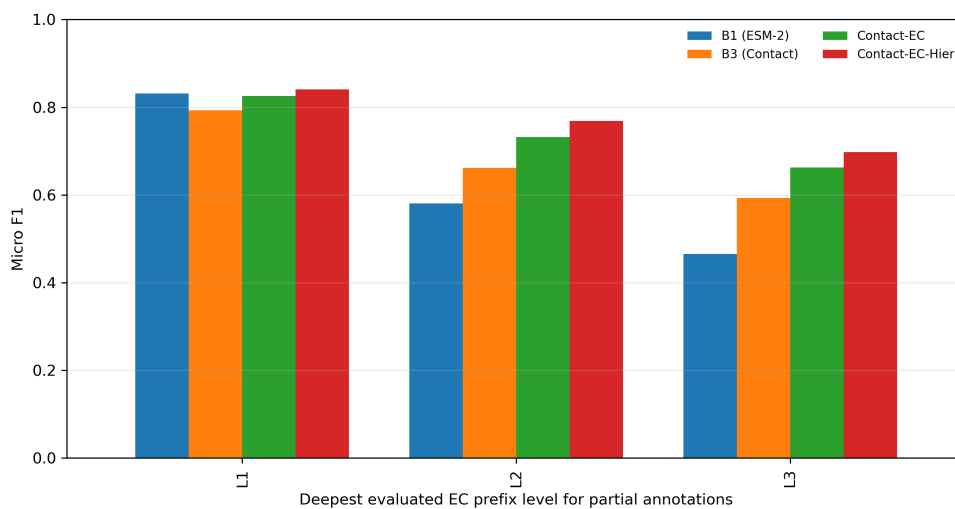


Figure 2: Hierarchical prefix micro F1 on partial-EC temporal proteins. The result shows that partial annotations should be handled as prefix-level supervision rather than assigned Level-4 failures.

Contact-EC (weighted F1 = 0.4763) exceeds EnzBert-learnnt (+1.3 pp) despite training on an earlier corpus with a  $4.5\times$  smaller backbone. Contact-EC-Hier (0.4325) falls 3.8 pp below EnzBert-learnnt; the gap to Contact-EC (+4.4 pp) reflects the flat FC head’s superior calibration on partial/novel-EC proteins under distribution shift.

Table 4: Case-wise HIT-EC vs. Contact-EC comparison on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. A case is counted as correct if the model recovers at least one true Level-4 EC label.

| True L1 | Both correct | HIT-EC only | Contact-EC only | Both wrong | Total |
|---------|--------------|-------------|-----------------|------------|-------|
| 1       | 12           | 5           | 1               | 0          | 18    |
| 2       | 13           | 8           | 1               | 3          | 25    |
| 3       | 29           | 11          | 0               | 2          | 42    |
| 4       | 5            | 20          | 0               | 7          | 32    |
| 5       | 4            | 1           | 0               | 2          | 7     |
| All     | 63           | 45          | 2               | 14         | 124   |

The case-wise comparison confirms that HIT-EC is the stronger temporal predictor in absolute terms. Contact-EC uniquely recovers two cases: A0A3G9HRC2, a multi-label oxidoreductase with EC 1.14.14.1 and EC 1.6.2.4, and A0A6C0WW38, where HIT-EC predicts the neighbouring methyltransferase class EC 2.1.1.159 instead of EC 2.1.1.160. Conversely, many HIT-EC-only successes are near misses for Contact-EC: for C0HLB8 (EC 3.1.1.4), Contact-EC assigns EC 3.1.1.4 the highest probability but below the fixed 0.5 threshold (0.4931). These cases support the main text’s interpretation that Contact-EC is useful for decomposing structure and calibration effects, whereas HIT-EC remains the stronger closed-set temporal classifier.

**EC Level-1 family analysis.** To add biological granularity, we recomputed temporal predictions for the ESM-2-only B1 model, the contact-only B3 model, and Contact-EC on the 124 fully evaluable temporal proteins, then stratified Level-4 micro F1 by true EC Level 1 family. Table 5 and Figure 3 show that Contact-EC yields the largest gains over ESM-2 in oxidoreductases (EC 1; +0.6270 micro F1) and hydrolases (EC 3; +0.2753), whereas lyases (EC 4) remain difficult for all Contact-EC variants despite HIT-EC’s stronger performance. These results suggest that the utility of contact-map fusion is enzyme-family dependent rather than uniform across EC space. Because several families contain few temporal examples, the analysis should be interpreted as hypothesis-generating rather than mechanistic evidence about active-site geometry.

Table 5: Level-1 family analysis on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. Values are Level-4 micro F1 within each true Level-1 family. EC 6 and EC 7 are absent from this evaluable subset.

| EC | Family          | N  | ESM-2  | Contact | Contact-EC | HIT-EC |
|----|-----------------|----|--------|---------|------------|--------|
| 1  | Oxidoreductases | 18 | 0.0909 | 0.2222  | 0.7179     | 0.9231 |
| 2  | Transferases    | 25 | 0.6667 | 0.6154  | 0.7059     | 0.8649 |
| 3  | Hydrolases      | 42 | 0.5085 | 0.4561  | 0.7838     | 0.9545 |
| 4  | Lyases          | 32 | 0.1923 | 0.0833  | 0.1695     | 0.6667 |
| 5  | Isomerases      | 7  | 0.3636 | 0.2500  | 0.6667     | 0.7143 |

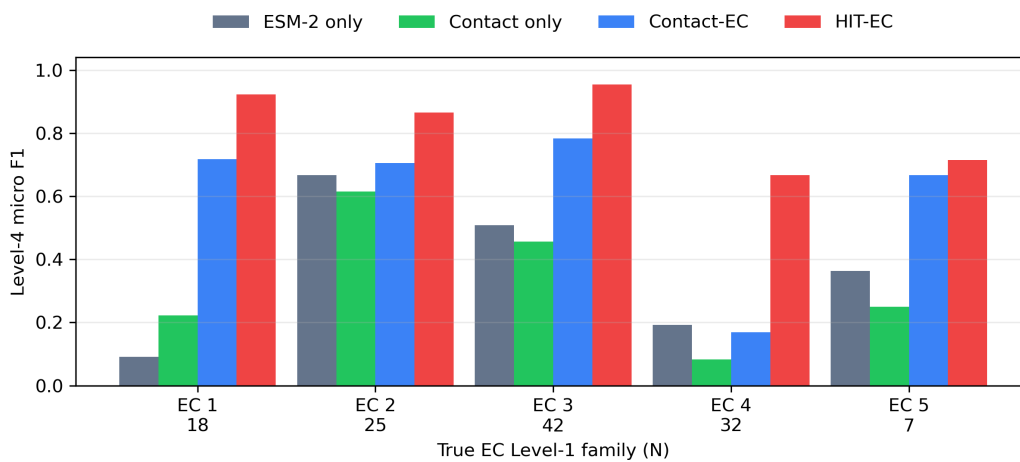


Figure 3: Temporal Level-4 micro F1 stratified by true EC Level-1 family. The largest Contact-EC gains occur in oxidoreductases and hydrolases, while lyases remain a failure mode for the current contact-fusion model.

Table 6: Contact-EC threshold sensitivity on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. The fixed threshold of 0.5 is used in the main benchmark table.

| Setting                 | Threshold | Micro F1 | Weighted F1 | Precision | Recall | Empty |
|-------------------------|-----------|----------|-------------|-----------|--------|-------|
| Fixed global            | 0.50      | 0.6032   | 0.5026      | 0.7677    | 0.4967 | 45    |
| Best global by micro F1 | 0.65      | 0.6167   | 0.4895      | 0.8506    | 0.4837 | 54    |
| 0.5 or top-1 if empty   | 0.50      | 0.5589   | 0.5413      | 0.5764    | 0.5425 | 0     |
| Top-1 only              | –         | 0.5126   | 0.4628      | 0.5726    | 0.4641 | 0     |

Threshold sensitivity analysis shows that the fixed 0.5 cutoff is not the sole explanation for Contact-EC’s temporal gap. Raising the global threshold to 0.65 slightly improves micro F1 from 0.6032 to 0.6167 by increasing precision, but reduces recall and weighted F1. Lower-threshold or top-1 fallback variants recover more labels but introduce additional false positives. This behaviour is consistent with a calibration–recall trade-off rather than a simple thresholding artifact.

## S5 External Benchmark Evaluation

Table 7: External benchmark results (per-protein F1, threshold = 0.5). CLEAN / CLEAN-Contact results from Ayres et al. [2024]. ¶: Contact-EC-ExpA on full SP-2026 (243K); New-392 pre-dates SP-2026, creating likely train/test overlap—interpret as upper bound.

| Method                              | Backbone               | Train data | New-392       | Price-149     |
|-------------------------------------|------------------------|------------|---------------|---------------|
| CLEAN-Contact [Ayres et al., 2024]  | ESM-2 3B               | SP-full    | <b>0.566</b>  | <b>0.525</b>  |
| CLEAN [Yu et al., 2023]             | ESM-2 3B               | SP-full    | 0.504         | 0.452         |
| ProteInfer [Sanderson et al., 2023] | CNN                    | UniRef90   | 0.309         | –             |
| B1 (ESM-2)                          | ESM-2 650M             | SP-2018    | 0.170         | 0.026         |
| B3 (Contact)                        | ResNet-50              | SP-2018    | 0.154         | 0.000         |
| Contact-EC (flat FC)                | ESM-2 650M + ResNet-50 | SP-2018    | <b>0.3717</b> | <b>0.1775</b> |
| Contact-EC-Hier                     | ESM-2 650M + ResNet-50 | SP-2018    | 0.358         | 0.096         |
| Contact-EC-ExpA ¶                   | ESM-2 650M + ResNet-50 | SP-2026    | 0.7680 ¶      | <b>0.2161</b> |

**New-392.** Contact-EC (SP-2018) achieves per-protein F1 = 0.3717, above ProteInfer (0.309) but below CLEAN (0.504) and CLEAN-Contact (0.566). The gap has two causes: (i) CLEAN-Contact uses ESM-2 3B (4.5× larger) and (ii) contrastive retrieval benefits from training-example similarity at inference. Contact-EC-ExpA (SP-2026) achieves 0.7680 but cannot be treated as a fair comparison since New-392 proteins pre-date 2026 and likely appear in the training corpus.

**Price-149.** Contact-EC (0.2500 micro F1) achieves the highest score among SP-2018 models, but the external Price-149 benchmark remains a major failure mode relative to CLEAN-Contact (0.525). To determine whether this gap is caused by missing inputs, we audited the raw 149 proteins, encoded SP-2018 evaluation subset, label vocabulary, and structure assets. Table 8 shows that input coverage is not the primary explanation: ESM-2 embeddings and nonzero contact maps are available for all 149 raw proteins, and 139 proteins have complete Level-4 EC labels contained in the SP-2018 vocabulary. Instead, the failure is concentrated at exact Level-4 specificity. In the hierarchical diagnostic, Contact-EC retains high coarse EC performance on Price-149 (Level 1 micro F1 = 0.9379; Level 2 = 0.7862; Level 3 = 0.7655), but Level 4 micro F1 falls to 0.0508 (Figure 4). The contact-only B3 model scores 0.0000 on the canonical Price-149 evaluation, and ESMFold-derived contact maps do not recover the gap (Contact-EC ESMFold-map micro F1 = 0.1218). We therefore interpret Price-149 as an external bacterial



RefSeq distribution-shift stress test: the model can often identify the broad enzyme family, but its current SP-2018 contact-fusion calibration does not transfer reliably to exact EC subclasses.

Table 8: Price-149 failure audit. Coverage is computed on the raw Price-149 file ( $N = 149$ ); canonical Level-4 model metrics use the encoded SP-2018 evaluation subset ( $N = 136$ ).

| Audit item                                               | Count/value | Interpretation                                  |
|----------------------------------------------------------|-------------|-------------------------------------------------|
| Raw Price-149 proteins                                   | 149         | external bacterial RefSeq benchmark             |
| Encoded SP-2018 subset                                   | 136         | canonical flat Level-4 evaluation subset        |
| Proteins with embeddings                                 | 149         | no missing ESM-2 input bottleneck               |
| Proteins with nonzero contact maps                       | 149         | no missing contact-map bottleneck               |
| Raw proteins with all EC labels in SP-2018 L4 vocabulary | 139         | moderate label-vocabulary coverage              |
| Unique raw Level-4 EC classes                            | 56          | external label diversity                        |
| Unique raw Level-4 EC classes outside SP-2018 vocabulary | 5           | limited closed-set mismatch                     |
| B1 ESM-2 Level-4 micro F1                                | 0.0324      | sequence-only transfer collapses                |
| B3 contact-only Level-4 micro F1                         | 0.0000      | contact-only transfer collapses                 |
| Contact-EC flat Level-4 micro F1                         | 0.2500      | fusion improves but remains below CLEAN-Contact |
| Contact-EC hierarchical Level-1 micro F1                 | 0.9379      | broad enzyme family often recovered             |
| Contact-EC hierarchical Level-4 micro F1                 | 0.0508      | exact subclass specificity fails                |

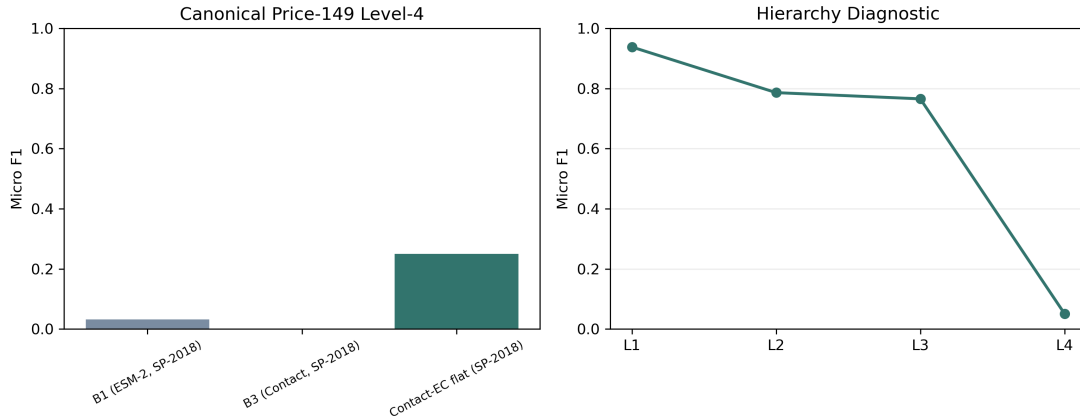


Figure 4: Price-149 failure analysis. Left: canonical Level-4 micro F1 for SP-2018 models. Right: hierarchical diagnostic for Contact-EC, showing that coarse EC levels remain predictable while exact Level-4 specificity collapses.

**Late probability fusion baselines.** To test whether Contact-EC’s gain could be reproduced by simple post-hoc combination of independently trained sequence-only and contact-only models, we evaluated probability-level fusion baselines using B1 and B3 predictions: mean posterior averaging, element-wise posterior maximum, and two fixed weighted averages. These baselines require no retraining and therefore provide a conservative diagnostic rather than a full architectural ablation. As shown in Table 9 and Figure 5, learned Contact-EC substantially outperforms all late-fusion variants on the 124 complete temporal proteins (0.6032 vs. 0.4673 for the best late-fusion baseline). The same pattern holds on Price-149, where late fusion remains near the weak ESM-only baseline while Contact-EC reaches 0.2500.

Thus the observed sequence–structure gain is not explained by naive posterior averaging or maximum pooling.

Table 9: Late probability-fusion baselines using independently trained B1 (ESM-2) and B3 (contact-only) models. Values are Level-4 micro F1 at threshold 0.5.

| Model                     | Swiss-Prot 2023 known EC ( $N = 124$ ) | Price-149 ( $N = 136$ ) |
|---------------------------|----------------------------------------|-------------------------|
| B1 (ESM-2)                | 0.4216                                 | 0.0324                  |
| B3 (Contact)              | 0.3646                                 | 0.0000                  |
| Late mean B1+B3           | 0.3770                                 | 0.0118                  |
| Late max B1+B3            | 0.4673                                 | 0.0312                  |
| Late weighted 0.7B1+0.3B3 | 0.4322                                 | 0.0231                  |
| Late weighted 0.3B1+0.7B3 | 0.3598                                 | 0.0000                  |
| Contact-EC learned fusion | <b>0.6032</b>                          | <b>0.2500</b>           |

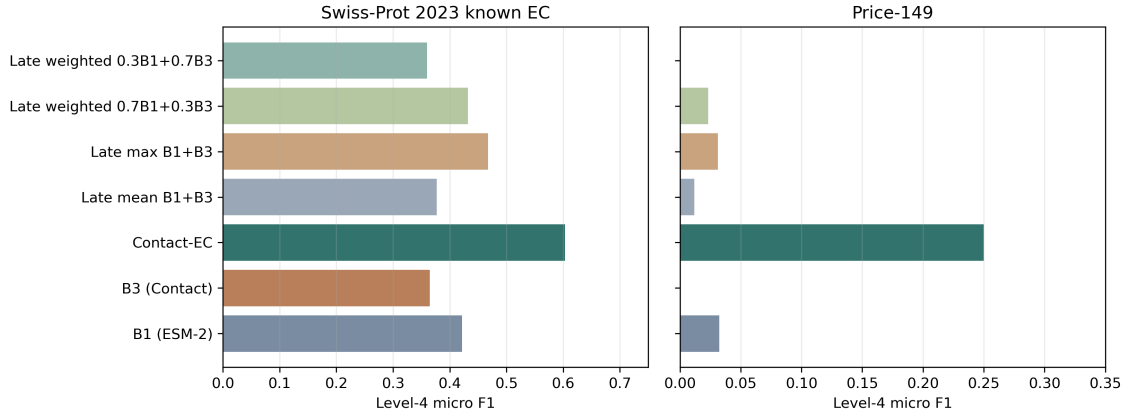


Figure 5: Late probability-fusion baselines. Learned Contact-EC exceeds simple B1/B3 posterior averaging, posterior maximum, and fixed weighted averages on both the temporal known-EC subset and Price-149.

## S6 Contact Map Channel Ablation

Table 10: Contact map channel perturbation (B3 checkpoint, random-split test,  $N = 22,018$  complete-L4 proteins). Inference-time zeroing of channels 1–2 is a pessimistic lower bound on the training-time 1-channel ablation.

| Input channels                                  | L4 Micro F1   | $\Delta$ vs. 3ch |
|-------------------------------------------------|---------------|------------------|
| 3-channel (all / short / long)                  | <b>0.8684</b> | —                |
| 1-channel <sup>†</sup> (ch0 only, ch1–2 zeroed) | 0.1187        | –74.97 pp        |

We encode contact maps as three channels—all contacts, short-range ( $|i-j| < 12$ ), and long-range ( $|i-j| \geq 12$ ). Inference-time zeroing of channels 1–2 drops L4 micro F1 from 0.8684 to 0.1187 (–74.97 pp), confirming the ResNet encoder exploits the channel decomposition to distinguish secondary from tertiary structure.

## S7 Grad-CAM Structural Interpretability

We apply Gradient-weighted Class Activation Mapping (Grad-CAM) [Selvaraju et al., 2017] to the ResNet-50 contact map branch to verify that Contact-EC encodes structurally meaningful contact patterns rather than superficial statistics.

For **Lysozyme C** (EC 3.2.1.17; UniProt C0HLB7), Grad-CAM highlights two discrete regions centred around residues 85–140 and 180–220, corresponding to the catalytic  $\beta$ -sheet strand and  $\alpha$ -helix cluster adjacent to the Glu35–Asp52 active-site dyad [Blake et al., 1967].

For **Phospholipase A1** (EC 3.1.1.32; UniProt P0DPT0), a single hot spot emerges around residues 110–175, co-localising with the serine–histidine–aspartate catalytic triad of the  $\alpha/\beta$  hydrolase fold [Schrag and Cygler, 1993].

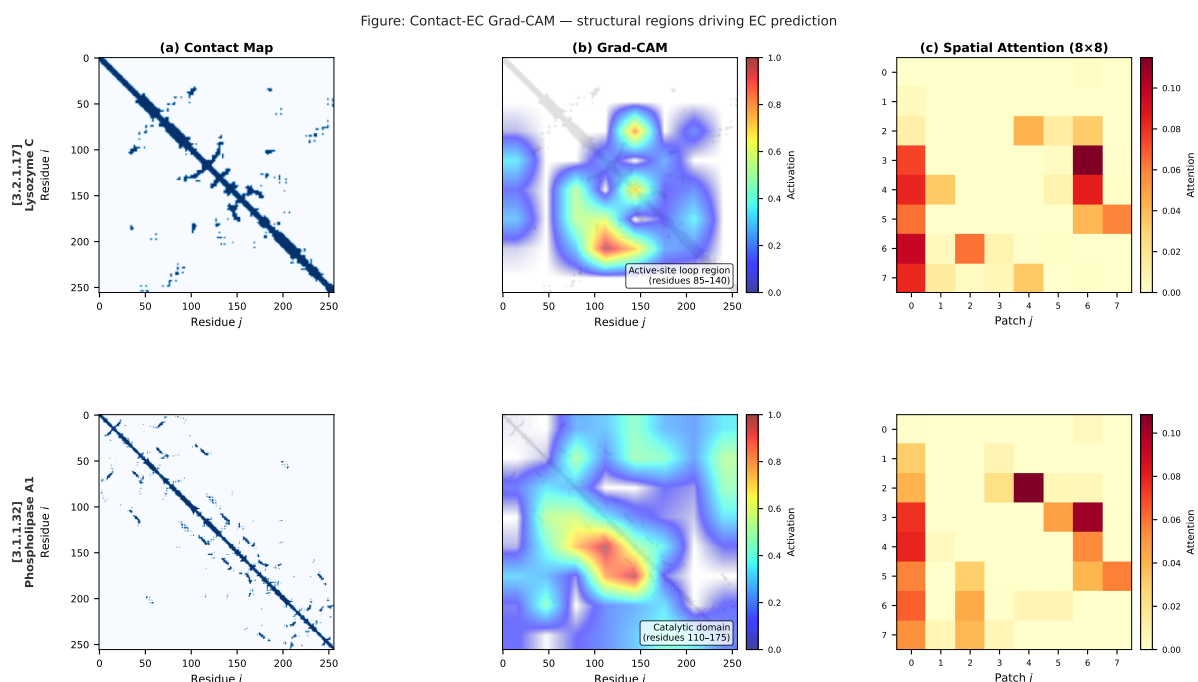


Figure 6: **Grad-CAM structural interpretability for Contact-EC.** Two correctly predicted test-set enzymes. (a) AlphaFold-derived contact map. (b) Grad-CAM heatmap overlaid on the contact map. (c) ResNet-50 spatial attention weights at the  $8 \times 8$  feature-map level. *Top:* Lysozyme C (EC 3.2.1.17, UniProt C0HLB7). *Bottom:* Phospholipase A1 (EC 3.1.1.32, UniProt P0DPT0).

## S8 Extended Discussion

**Comparison to CLEAN-Contact.** CLEAN-Contact uses ESM-2 3B (2,560-dimensional) with contrastive kNN inference and full Swiss-Prot training data. On New-392, Contact-EC (0.3717, SP-2018) trails CLEAN-Contact (0.566) due to backbone scale (3B vs. 650M) and training recency. Contact-EC-ExpA (SP-2026, 0.7680) surpasses CLEAN-Contact but with potential train/test overlap. The gap on Price-149 reflects the OOD nature of bacterial RefSeq sequences and exact Level-4 calibration under structure-source shift rather than a simple missing-input problem.

**Paper significance and scope.** The current results demonstrate that structural contact maps provide complementary and synergistic information alongside PLM embeddings, empirically supported on rigorous OOD evaluation. The structural parity finding ( $B3 \approx B1$  on hard OOD validation) is, to our knowledge, a novel empirical observation: prior structure-aware EC methods either use contact maps with much larger sequence models or treat structure as a minor addition.

**Threshold calibration and temporal generalization.** At inference, we apply a fixed sigmoid threshold of 0.5 for Level-4 EC predictions. The sensitivity analysis in Table 6 indicates that this cutoff is reasonably stable for micro F1 but trades precision against recall. The gap in Rare-EC F1 between Contact-EC (0.4219) and Contact-EC-Hier (0.3621) further suggests that the hierarchical head suppresses rare-class outputs through cross-level attention. Future work should explore class-adaptive thresholding and calibrated decision rules to improve rare-class recall without increasing false positive assignments.

**3B backbone scaling.** Contact-EC-3B (ESM-2 3B backbone, flat FC head, Phase 1 only) achieves micro F1=0.6316 on the temporal test, +2.8 pp over the 650M model. The modest gain for a  $4.5\times$  parameter increase suggests the model is data-constrained rather than backbone-constrained at the current training scale.

## S9 Reproducibility and Audit Checks

All reported splits were audited for accession overlap, exact sequence overlap, label coverage, multi-label cardinality, and contact-map availability. The audit scripts generate machine-readable CSV summaries used to construct the final result tables. Random-split, cluster-split, EC-Bench-style, and external benchmark results are labeled separately to avoid mixing evaluation protocols.

Table 11: Final temporal leakage audit for the 124 fully evaluable Swiss-Prot 2023-01 proteins. Accession and exact-sequence overlap are necessary checks but do not establish homolog- or fold-disjointness.

| Training corpus              | Train $N$ | Test $N$ | Acc. overlap | Seq. overlap | Train L4 |
|------------------------------|-----------|----------|--------------|--------------|----------|
| SP-2018 EC-Bench train       | 201,865   | 124      | 0            | 0            | 4,284    |
| ExpA recent Swiss-Prot train | 243,198   | 124      | 0            | 0            | 5,145    |

**Sequence-similarity stratification of temporal cases.** To test whether temporal performance is concentrated in proteins with close training-set neighbours, we joined the final case-wise temporal comparison ( $N = 124$ ) with the available EC-Bench test-vs-train similarity audit. This similarity file covers 101 of the 124 fully evaluable temporal proteins; the remaining 23 proteins are therefore excluded from the stratified analysis. Table 12 and Figure 7 show that Contact-EC performance is substantially lower when no recorded sequence similarity to the training set is available, and increases in the 0.60–0.90 identity regime. This supports the manuscript’s conservative interpretation: temporal EC performance is sensitive to train-test relatedness, and sequence-disjoint validation does not by itself establish fold-disjoint structural generalization.

Table 12: Temporal performance stratified by maximum recorded train-set sequence identity for the 101 temporal proteins with available EC-Bench similarity metadata. This is a sequence-similarity sensitivity audit, not a structural fold-disjoint evaluation.

| Max train identity | $N$ | Contact-EC F1 | HIT-EC F1 | Contact hit | HIT-EC hit |
|--------------------|-----|---------------|-----------|-------------|------------|
| 0.00               | 32  | 0.3673        | 0.8125    | 0.2812      | 0.8125     |
| (0.30, 0.60]       | 36  | 0.5902        | 0.8108    | 0.5000      | 0.8333     |
| (0.60, 0.90]       | 27  | 0.8889        | 0.9643    | 0.8889      | 1.0000     |
| >0.90              | 6   | 0.8000        | 0.8333    | 0.6667      | 0.8333     |

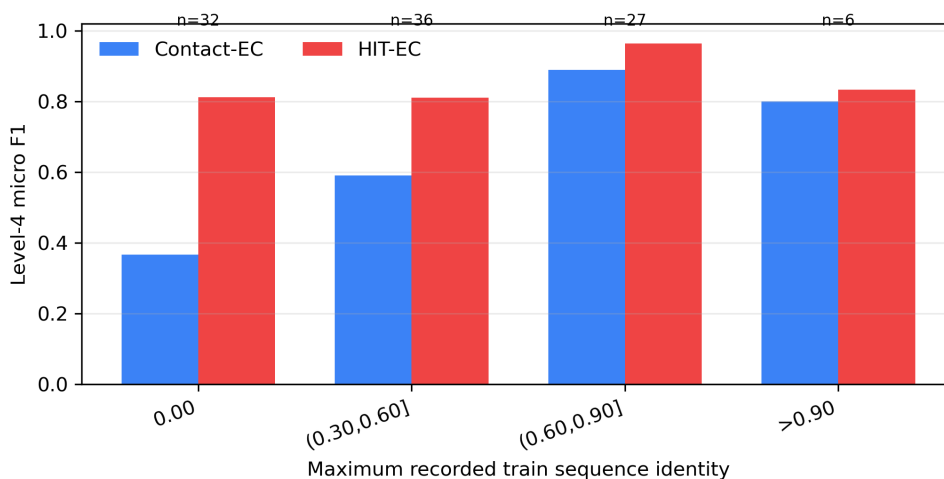


Figure 7: Level-4 micro F1 by maximum recorded train-set sequence identity on the 101 temporal proteins with available EC-Bench similarity metadata. The result motivates future Foldseek/TM-align or CATH/SCOP fold-disjoint testing.

**Contact-pair nearest-neighbor audit.** Because sequence-disjoint evaluation does not guarantee structural fold-disjointness, we also performed a model-proximal relatedness audit using the existing contact-pair ESM representation. For each protein, we summarized the contact-pair tensor by concatenating the mean and standard deviation across the sampled residue pairs and computed cosine similarity to all processed SP-2018 training proteins with available contact-pair embeddings. This analysis is not a Foldseek, TM-align, CATH, or SCOP fold-disjoint benchmark. Instead, it asks whether temporal test proteins have close neighbours in the same representation space used by the contact branch. Table 13 and Figure 8 show that all 113 evaluable temporal known-EC proteins have a nearest training neighbour with contact-pair cosine similarity above 0.90. The nearest neighbour shares the same Level-4 EC label in 59.3% of cases. We therefore avoid interpreting the temporal results as evidence of fold-disjoint structural generalization; a true claim of that type requires Foldseek/TM-align or CATH/SCOP clustering followed by retraining and evaluation on the resulting fold-disjoint split.

Table 13: Contact-pair nearest-neighbor audit for temporal known-EC proteins. Cosine similarity is computed on mean–standard deviation summaries of existing contact-pair ESM tensors. This is a representation-neighbour audit, not a structural fold-disjoint benchmark.

| Quantity                                                | Value           |
|---------------------------------------------------------|-----------------|
| Temporal known-EC proteins with contact-pair embeddings | 113/124         |
| SP-2018 training proteins with contact-pair embeddings  | 212,863/224,294 |
| Median nearest-neighbour cosine                         | 0.9852          |
| Mean nearest-neighbour cosine                           | 0.9845          |
| Temporal proteins with nearest cosine >0.90             | 113/113         |
| Nearest neighbour same Level-1 EC                       | 87.6%           |
| Nearest neighbour same Level-2 EC                       | 82.3%           |
| Nearest neighbour same Level-3 EC                       | 82.3%           |
| Nearest neighbour same Level-4 EC                       | 59.3%           |

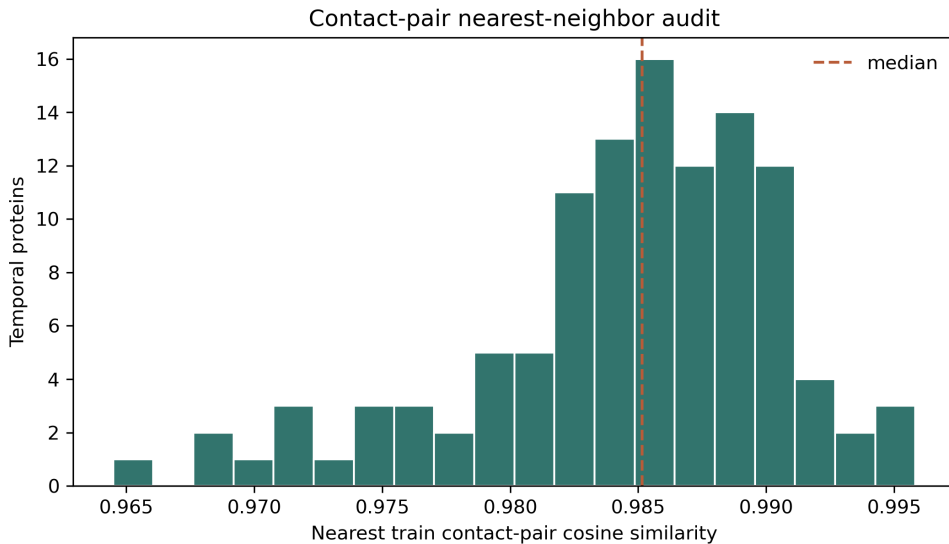


Figure 8: Distribution of nearest SP-2018 training-set cosine similarity for the 113 temporal known-EC proteins with available contact-pair embeddings. The high nearest-neighbour similarities indicate that the current temporal evaluation should not be treated as fold-disjoint structural generalization.

Table 14: Interpretation of each evaluation protocol. The same numerical metric can answer different scientific questions depending on split construction and label coverage.

| Protocol                           | Primary question                                                                 | Main caveat                                                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random Swiss-Prot split            | In-distribution fitting and calibration                                          | Contains exact sequence duplicates across partitions; not evidence of temporal or sequence-disjoint generalization.                                            |
| MMSeqs2 cluster split              | Generalization to sequence-disjoint proteins                                     | Valid only when the evaluated checkpoint is trained on the corresponding cluster-train split.                                                                  |
| EC-Bench val_hard                  | Controlled hard validation at $\leq 30\%$ sequence identity                      | Still a validation protocol; temporal deployment claims require held-out future annotations.                                                                   |
| Swiss-Prot 2023-01 known-EC subset | Closed-set recognition of temporally new proteins whose labels are in vocabulary | Only 124/468 proteins are fully evaluable at Level 4.                                                                                                          |
| Swiss-Prot 2023-01 full set        | Sensitivity to temporal label shift and incomplete annotations                   | Operational coverage audit only; partial labels are not fully Level-4 scorable, and novel labels are outside the closed-set vocabulary.                        |
| New-392 / Price-149                | External stress tests from CLEAN-Contact                                         | New-392 overlaps with current Swiss-Prot; Price-149 is an external bacterial RefSeq stress test with strong Level-4 specificity loss despite available inputs. |

These protocol distinctions are used throughout the manuscript to avoid treating all F1 values as interchangeable. In particular, the Bioinformatics-relevant claim is not that Contact-EC is the strongest available EC predictor in absolute terms. Rather, Contact-EC provides a controlled sequence–structure framework for separating representation effects from temporal cutoff, label-vocabulary shift, sequence overlap, and structure availability.

**Canonical result selection.** When multiple evaluation logs exist for the same checkpoint, we use the full multi-label evaluation on the full evaluable temporal set ( $N = 124$ ) as canonical. Earlier  $N = 101$  evaluations are retained only as historical partial runs and are not used in the main conclusions. For Contact-EC-3B, the canonical temporal result is micro F1 = 0.6316 from the full  $N = 124$  multi-label evaluation; auxiliary argmax-only and partial-set evaluations are excluded from the main result table. For B3 contact-only hard validation, we use the completed resumed checkpoint evaluation (micro F1 = 0.7650, weighted F1 = 0.6788, macro F1 = 0.1335); an earlier interrupted training log reached a higher validation micro F1 before termination but is not used as the paper source because the completed run provides the auditable checkpoint/result pair.

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